

5 *Mycobacterium tuberculosis* (*Mtb*) and human immunodeficiency virus (HIV) are two pathogens that infect people.

When people infected with either pathogen become infected with the other pathogen the disease is worse.

(a) Explain why HIV infection causes tuberculosis (TB) to be worse.

(4)

(b) In one year, it was estimated that 33.4 million people were infected with HIV. It was estimated that 30% of these people were also infected with *Mtb*.

(i) Suggest **two** reasons why these figures are only estimates.

(2)

(ii) Calculate the number of people with both infections.

Give your answer in standard form.

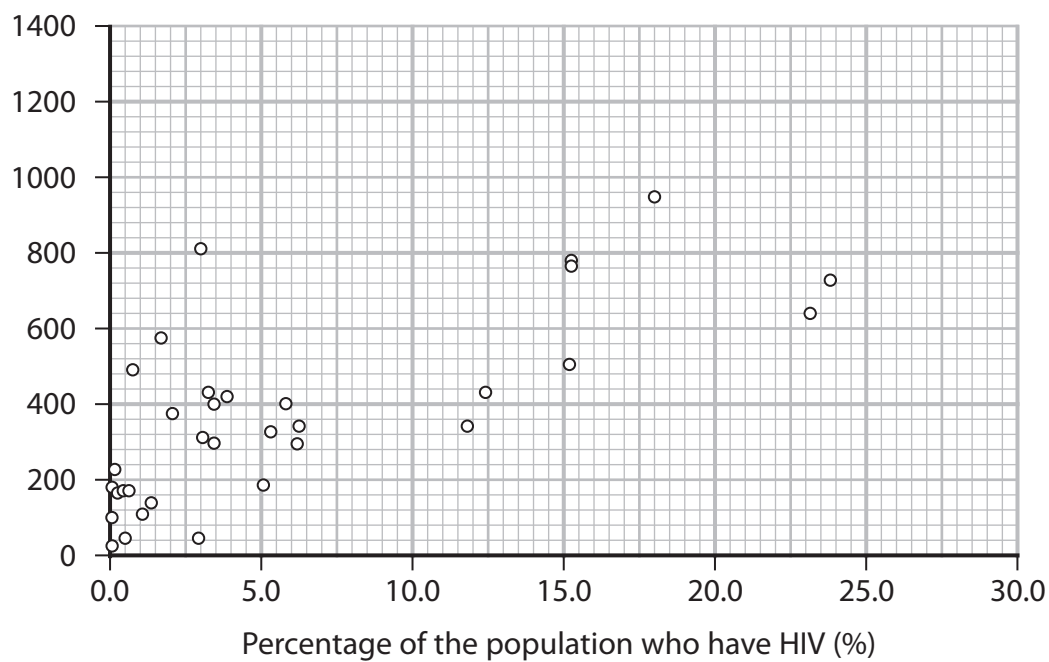
(1)

Answer

(c) The graph shows the number of people who have TB and the percentage of the population who have HIV.

Each plotted point represents data from a different country.

Number of people who have TB
per 100 000 population



- (i) Comment on the relationship between the number of people who have TB and the percentage of people who have HIV.

(2)

- (ii) Describe how the strength of the relationship between the number of people who have TB and the percentage of people who have HIV could be determined.

(3)

(Total for Question 5 = 12 marks)